

NICOLÁS CASTRO-PERDOMO

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Address: 1001 E 10th St, 47405 Bloomington, Indiana, USA.

EDUCATION

Indiana University

PhD. in Geological Sciences

GPA: 3.9/4.0

Bloomington, United States

August 2022 - Present

King Abdullah University of Science and Technology

M.S. in Earth Science and Engineering

GPA: 3.85/4.0

Thuwal, Saudi Arabia

December, 2019

Universidad Nacional de Colombia

B.S. in Geology · Honors Degree

GPA: 4.4/5.0

Bogotá, Colombia

September, 2017

RESEARCH EXPERIENCE

Graduate Student Researcher

Indiana University

Advisor: Kaj Johnson

Bloomington, USA

August, 2022 - Present

Visiting Research Intern

King Abdullah University of Science and Technology

Advisor: Sigurjón Jónsson

Thuwal, Saudi Arabia

March, 2022 - August 2022

Graduate Student Researcher

King Abdullah University of Science and Technology

Advisor: Sigurjón Jónsson

Thuwal, Saudi Arabia

August, 2017 - December, 2019

Visiting Research Intern

EOST, Université de Strasbourg

Advisor: Frédéric Masson

Strasbourg, France

June, 2018 - July, 2018

Visiting Research Intern

Cornell University

Advisor: Larry D. Brown

Ithaca, United States

January, 2016 - July, 2016

PUBLICATIONS

1. **Nicolás Castro-Perdomo**, Sigurjón Jónsson, Yann Klinger, Frederic Masson, Thorsten Wolfgang Becker, and Kaj Johnson, **2025**. Strain rates along the Alpine-Himalayan belt from a comprehensive GNSS velocity field. *JGR: Solid Earth*. DOI: [10.1029/2025JB031738](https://doi.org/10.1029/2025JB031738)
2. Xing Li, Sigurjón Jónsson, Shaozhuo Liu, and Zhangfeng Ma, **Nicolás Castro-Perdomo**, Simone Cesca, Frédéric Masson, and Yann Klinger, **2024**. Resolving the slip-rate inconsistency of the northern Dead Sea fault. *Science Advances*. DOI: [10.1126/sciadv.adj8408](https://doi.org/10.1126/sciadv.adj8408)
3. **Nicolás Castro-Perdomo**, Renier Viltres, Frédéric Masson, and Yann Klinger, Shaozhuo Liu, Maher Dhahry, Patrice Ulrich, Jean-Daniel Bernard, Rémi Matrau, Abdulaziz Alothman, Hani Zahran, Robert Reilinger, P Martin Mai, and Sigurjón Jónsson, **2022**. Interseismic Deformation in the Gulf of Aqaba from GPS Measurements. *Geophysical Journal International*. DOI: [10.1093/gji/ggab353](https://doi.org/10.1093/gji/ggab353)

MANUSCRIPTS IN PREPARATION

4. **Nicolás Castro-Perdomo** and Kaj Johnson, **in prep.** Satellite Geodesy Reveals Substantial Aseismic Off-fault Deformation in California.
5. **Nicolás Castro-Perdomo** and Kaj Johnson, **in prep.** Fault Complexity Controls Deformation Partitioning Between Faults and the Surrounding Crust

SOFTWARE

1. **Nicolás Castro-Perdomo (2024)**. FICORO_GNSS: An open-source Python software package for filtering, combining and rotating GNSS velocity fields. DOI: [10.5281/zenodo.13921188](https://doi.org/10.5281/zenodo.13921188).

HONORS AND AWARDS

NSF travel award (administered through Stanford University) Numerical Modeling of Earthquake Motions Workshop	Smolenice, Slovakia <i>June, 2024</i>
Graduation with honors (highest distinction) Universidad Nacional de Colombia	Bogotá, Colombia <i>September, 2017</i>
Valedictorian - Class rank: 1/40 Universidad Nacional de Colombia	Bogotá, Colombia <i>September, 2017</i>
First place award 6th KAUST Annual International Undergraduate Research Poster Competition	Thuwal, Saudi Arabia <i>January, 2017</i>
Nexo Global - Colciencias scholarship Fully funded research internship at Cornell University	Ithaca, United States <i>January-July, 2016</i>
Dean's list - Tuition fee waiver Universidad Nacional de Colombia	Bogotá, Colombia <i>2013 - 2017</i>

INVITED PRESENTATIONS AND KEYNOTES

Earth Sciences New Zealand (Formerly GNS Science) "Slip Deficit Rate Inversions in New Zealand from Geodetic Data"	Online <i>October, 2025</i>
COMET webinar series "A Comprehensive Model of Crustal Deformation in the Mediterranean"	Online <i>November, 2023</i>

SUMMER COURSES

International School of Physics "Enrico Fermi" Italian Physical Society Course 202: Mechanics of Earthquake Faulting	Varenna, Italy <i>July 2018 (1 week)</i>
Workshop on Mechanics of the Earthquake Cycle International Centre for Theoretical Physics (ICTP) Course SMR-3885	Trieste, Italy <i>October 2023 (2 weeks)</i>
Workshop on Numerical Modeling of Earthquake Motions Slovak Academy of Sciences NMEM 2024: Waves and Ruptures	Smolenice, Slovakia <i>June 2024 (1 week)</i>

VOLUNTEER AND PROFESSIONAL SERVICE

Student Representative

American Geophysical Union (AGU). Geodesy section, 2025-2027

Peer Reviewer

JGR: Solid Earth, Tectonophysics, GJI

SKILLS

Computer Skills

Programming: Python, MATLAB, Bash, Julia (learning)

GNSS data analysis: GAMIT/GLOBK, CATS, HECTOR

Others: HPC-SLURM, PSGRN/PSCMP, TDEFNODE, GMT, L^AT_EX

OS: GNU/Linux, MacOS, Windows

Language Skills

Spanish: First language

English: C1 - IELTS 7.0

FIELDWORK EXPERIENCE

Geodetic survey

Surveyed 30 campaign GNSS stations

Gulf of Aqaba, Saudi Arabia

March-April 2022 (2 weeks)

Drone survey

Collaborated in the acquisition of drone imagery

Húsavík, Iceland

July 2018 (1 week)

CorNET seismic network

Collected data from 3 seismic stations

Ithaca, United States

June, 2016 (2 days)